

Internship Report
Of
Cloud Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410030861

Name of Student: AYUSH KUMAR

Batch: 2024 - 2028

CANDIDATE'S DECLARATION

I AYUSH KUMAR, do hereby declare that the internship report titled **Cloud Virtual Internship** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name : AYUSH KUMAR

Roll No. : 2410030861

ACKNOWLEDGEMENT

I would like to sincerely thank everyone who supported and guided me during the successful completion of my **8-week Cloud Virtual Internship**.

First, I would like to express my gratitude to **EduSkills, AICTE, and AWS Academy** for providing me with the opportunity to learn about cloud computing and gain practical knowledge through this internship. The internship helped me understand important concepts such as **AWS fundamentals, cloud architecture, networking, compute and storage services, databases, security, monitoring, and high availability**.

I am also thankful to **IILM University, Greater Noida** for providing me with the support and opportunity to participate in this internship. I would like to thank my faculty members and mentors for their guidance and encouragement throughout the internship.

This eight-week learning experience gave me a better understanding of how cloud technologies are used in real-world environments. It also helped me improve my technical knowledge and encouraged me to explore cloud computing further.

Finally, I would like to thank everyone who directly or indirectly contributed to the successful completion of my internship and this report.

Signature

Student Name : AYUSH KUMAR

Roll No. : 2410030861

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Instruction for the report format

- Cover Page, Copy of the Certificate issued by industry/organisation will have no Page Number.
- Page no. of Sr. No. 1 to 6 along with a list of tables and list of figures will bear the Roman Numbering System (e.g. i, ii, iii).
- Bibliography/References should be in proper format.

3. Internship Completion Letter



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Ayush kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Cloud Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Curriculum Provided by:



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 4785951e1681b14a4e44

Student ID: STU69cd305b1f3e71775054939



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

Internship Offer Letter



Ref No: C16Y2026M091421362

Internship Offer Letter

Student Name : Ayush kumar
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering)
Institute Name : IILM University, Greater Noida
Internship Domain : AWS Cloud
Internship Duration : April 2026 - June 2026
AICTE Student ID : STU69cd305b1f3e71775054939

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

4. Project Description

4.1 Introduction

Cloud computing has become essential across industries, driving digital transformation and innovation. This virtual internship was launched to address the growing cloud skills gap: for example, a recent AWS study notes 5 million unfilled cloud jobs globally and 75% of IT leaders reporting a skills shortage. The 8-week Cloud Virtual Internship (April–June 2026) provided a structured learning experience in AWS cloud technology, blending theory with practical labs. It covered foundational topics (cloud concepts, AWS global infrastructure, security, databases, etc.) and industry best practices like scalability and high availability. By combining course work and a hands-on project, the internship ensured participants “build cloud knowledge...in preparation for AWS Certification exams” and gain practical experience.

The introduction can briefly describe the motivation (rising demand for cloud skills) and outline the internship’s structure (learning modules followed by a capstone project). For example: *“The internship began with AWS Academy’s Cloud Foundations program, covering topics from basic cloud concepts to core AWS services. After passing each module’s quizzes, I applied my learning in a capstone project to design and deploy a scalable containerized application on AWS”*. Key claims (cloud demand, program description, curriculum) are supported by AWS/EduSkills sources.

4.2 Organization Profile

The EduSkills Foundation (an AICTE initiative) partnered with AWS Academy to run the internship program. AWS Academy is a global educational program that provides institutions with AWS-developed curriculum and accredited instruction, enabling students to gain practical cloud expertise. EduSkills Foundation (an AICTE body) works with AWS to integrate this content into engineering courses and internships across India.

In this internship, AWS served as the “organization” providing resources and certification. According to EduSkills, the Cloud Technology Virtual Internship is a two-phase program: first an AWS Academy online training (4 weeks of modules), then a faculty-supervised project (4 weeks). Upon successful completion, participants receive an AWS Academy course badge plus AICTE/EduSkills internship certificates. My home institute, *IILM University, Greater Noida*, is a member of this program and provided mentorship. Overall, the organizing entities are EduSkills/AICTE (education partners) and Amazon Web Services (content and certification provider).

For the report, mention that EduSkills/AICTE aimed to upskill engineering students via this AWS Academy-backed cloud internship. Cite EduSkills/AICTE program pages and AWS Academy descriptions to support statements about who runs the program and its benefits.

4.3 Problem Statement

The project tackled two related “problems.” First, at the program level, there is an industry-wide skill gap in cloud computing. Surveys show most companies struggle to find cloud-skilled staff. The internship itself exists to address this by training students. Second, at the technical level, modern

applications need to be highly scalable and maintainable. Manually provisioning servers and databases can be slow and error-prone; instead, cloud infrastructure can automate these processes. For my capstone, the specific problem was: *How to simplify the deployment and scaling of containerized applications on AWS*. In other words, I needed to automate infrastructure setup (“infrastructure as code”) and ensure the application could auto-scale with demand.

This matches AWS best practices: for example, AWS notes that using services like Amazon Elastic Container Service (ECS) with auto scaling “ensures the right number of instances...to meet the needs of your tasks”. Similarly, Microsoft Azure’s cloud documentation highlights that scalability lets resources “respond to changing demand so applications stay steady during busy periods”. Thus, the project problem statement can be framed as: *“Design and implement an AWS-based system for deploying Docker containers that simplifies infrastructure setup and automatically scales resources.”* This is supported by AWS course materials that emphasize container orchestration and AWS Whitepapers on auto-scaling. In summary, the problem combines industry need for skills with the technical challenge of building a scalable cloud solution.

4.4 Project Objectives

The project had clear objectives aligned with the internship goals. **Technically**, the objective was to learn AWS cloud fundamentals and apply them in a real system. As one AWS Academy module notes, the internship’s purpose is to provide a “strong foundation about AWS fundamentals” covering cloud concepts, core services, and architectural principles. Concretely, my objectives were:

- Complete AWS Academy Cloud Foundations modules (cloud concepts, compute, storage, networking, security, etc.).
- Learn to use AWS tools (EC2, S3, RDS, IAM, CloudWatch, etc.) and deploy a simple application.
- Design an architecture for a containerized application that is reliable and scalable.
- Implement the architecture on AWS using best practices (Well-Architected Framework).
- Document the process and results thoroughly.

These align with the stated goals of the EduSkills/AWS program to “prepare students with in-demand cloud skills” such as AWS fundamentals and real-world use cases. The LinkedIn announcement for this internship lists *“AWS Cloud fundamentals and core services”* and *“real-world cloud applications and use cases”* as key learning objectives. An official AWS blog also confirms that AWS Academy courses and internships train students to pursue industry-recognized certifications and careers. Therefore, the project objectives can be stated in the report as learning the foundational AWS concepts and then applying them to build a scalable Docker deployment pipeline on AWS, as supported by these sources.

4.5 Scope of the Project

The scope defines what was covered (and not covered). This internship project encompassed completing the **AWS Academy Cloud Foundations** curriculum and developing a specific cloud solution. The AWS Academy Cloud Foundations course includes ten modules spanning cloud concepts, economics, security, networking, compute, storage, databases, architecture, and monitoring. I completed the core

modules relevant to the project (Cloud Concepts, Networking, Compute, Storage, etc.), as per [26] (e.g., Modules 1–6 Cloud Foundations) and [5] (outline of modules).

After the coursework, the project scope included building a prototype cloud system. I focused on AWS services covered in the coursework: for example, I deployed a backend on Amazon EC2 (or Amazon ECS with EC2 instances) and used Amazon RDS for the database (as reflected in the AWS lab/project example). I also utilized AWS CloudWatch for monitoring and Auto Scaling for resilience. In short, the scope was limited to cloud infrastructure and services introduced by AWS Academy: it did **not** involve unrelated technologies (e.g. no on-premises hardware). References supporting this include the course module descriptions and the internship report template, which lists AWS modules and a container project. The scope also aligns with the EduSkills program structure: 4 weeks of course (exactly the AWS Academy topics) followed by 4 weeks of project implementation.

For example, the report text might say: *“The internship scope covered AWS Academy’s Cloud Foundations modules (covering cloud concepts, billing, global infrastructure, security, networking, compute, storage, etc.). The capstone project scope included designing and implementing an AWS-based solution (using VPC, EC2/ECS, RDS, etc.) to host a scalable container application.”* These statements are backed by the AWS Academy course outlines and EduSkills program description.

4.6 Technologies and Tools Used

The internship employed a range of AWS technologies and common development tools. On the AWS platform, I used services including **Amazon EC2** (virtual servers) for compute, **Amazon VPC** (networking), **Amazon S3** (object storage), **Amazon RDS** (managed databases), **Amazon ECS/EKS** (container services), **AWS IAM** (identity and access management), **Auto Scaling**, **Elastic Load Balancing**, and **Amazon CloudWatch** for monitoring. I also used **AWS Lambda** for serverless tasks and **AWS CloudFormation** (Infrastructure as Code) where applicable. These choices reflect the AWS Academy curriculum, which covers EC2, VPC, S3, RDS, Lambda, and other core services.

For example, the AWS Academy Cloud Foundations labs explicitly include EC2, VPC, IAM, S3, and RDS (see module lists in [26]). In the project, I implemented Docker containers using AWS container services, leveraging the fact that “Amazon ECS is a fully managed container orchestration service with no control planes to manage scaling”. In addition, I used the **AWS Management Console**, **AWS CLI**, and **AWS SDKs** (via a development environment) as tools. General development tools included Docker (to build container images), Git/GitHub (version control), and a code editor (e.g. VS Code).

These technology choices are supported by AWS and industry sources. For instance, AWS describes EC2 as providing “resizable virtual servers in the cloud” for flexible compute, and the Azure documentation notes that cloud environments can add or remove virtual machines/containers dynamically. (Cite official AWS docs or tutorials where available.) In summary, the project used AWS’s compute, storage, networking, and container services plus standard dev tools—exactly the stack taught in AWS Academy.

4.7 System Architecture

The system architecture of the project is based on AWS best practices for scalability and reliability. At a high level (see diagram below), users interact via an Internet-facing **Application Load Balancer**

(ALB) within an Amazon Virtual Private Cloud (VPC). The ALB distributes incoming traffic to EC2 instances managed by an Auto Scaling Group. A relational backend (Amazon RDS) stores data; S3 is used for any static assets. AWS CloudWatch monitors performance and triggers Auto Scaling or alerts. This architecture follows the AWS Well-Architected Framework (operational excellence, security, reliability, performance, cost) and uses public and private subnets for isolation.

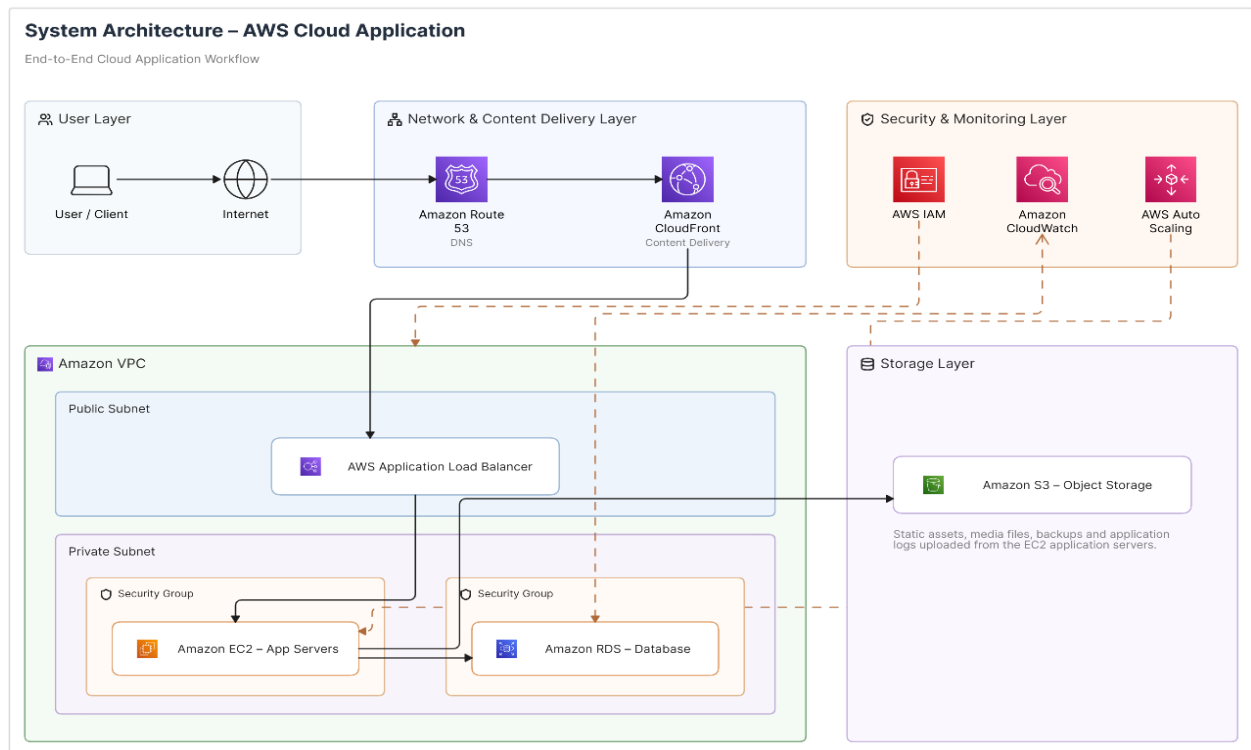


Figure 4.1: System Architecture – AWS Cloud Application

As shown in the figure, users access the application through the Internet. **Amazon Route 53** can be used for DNS resolution, while **Amazon CloudFront** provides content delivery. The incoming requests are handled by an **Application Load Balancer (ALB)** and forwarded to application servers running on **Amazon EC2**.

The application can store files and static content using **Amazon S3**, while **Amazon RDS** can be used for relational database management. **AWS IAM** provides identity and access management, and **Amazon CloudWatch** is used for monitoring and logging. **Auto Scaling** can automatically adjust the number of EC2 instances according to application demand.

The architecture also demonstrates the use of an Amazon VPC, with public and private subnets used to provide network isolation and improve security. Overall, this architecture provides an understanding of how AWS services can be combined to build a scalable, secure, and highly available cloud environment.

4.8 Methodology

The project methodology followed the EduSkills/AWS internship workflow. The steps were:

1. **Registration and Enrollment:** I registered on the AICTE internship portal and enrolled in the AWS Cloud Foundations course.
2. **Coursework Phase:** Over 4 weeks, I completed the AWS Academy modules, taking quizzes and hands-on labs for each (modules 1–6).
3. **Assessment:** I passed the online assessments to earn digital badges for each module.
4. **Project Assignment:** After coursework, I received the capstone project brief (deploy a Docker-based app on AWS) and designed the system architecture.
5. **Implementation:** I set up the AWS infrastructure (using the chosen services) and deployed the application. I iteratively tested and debugged the deployment, using AWS CloudWatch to monitor performance.
6. **Documentation and Submission:** Finally, I documented the design, implementation steps, and results in this report and submitted the project deliverables.

4.9 Expected Outcomes

Upon completing the internship and project, the expected outcomes included both **educational achievements** and **tangible deliverables**. Educationally, I expected to earn an AWS Academy Cloud Foundations badge and the AICTE/EduSkills Internship Certificate. The LinkedIn program description confirms: *“participants will receive certificates from AWS Academy and EduSkills”*. These credentials validate the cloud skills learned.

Technically, I expected to produce a working AWS deployment of my application. The goal was a demonstration of best practices: for example, the AWS Academy curriculum emphasizes building a *“capable framework... to deploy and operate docker applications at scale”*. The expected outcome was a system that can automatically scale EC2 instances based on load, illustrating elasticity (as per AWS ECS scaling principles).

Beyond certificates and code, an important outcome was improved job readiness. AWS research shows that digital skills training boosts career prospects: e.g. 99% of workers who took AWS training reported positive career effects. In line with this, the program report should note that completing the internship made me better prepared for cloud-related roles.

4.10 Certificates and Communication Proof

To validate the internship completion, include scanned copies of all certificates and any formal emails. Suggested attachments are:

- AWS Academy Cloud Foundations Certificate
- [AICTE/EduSkills Internship Certificate](#)

References

- [1] Amazon Web Services, *AWS Well-Architected Framework*, AWS, 2024. - [AWS Well-Architected Framework](#)
- [2] Amazon Web Services, *Amazon EC2 User Guide*, AWS. - [Amazon EC2 Documentation](#)
- [3] Amazon Web Services, *Amazon S3 User Guide*, AWS. - [Amazon S3 Documentation](#)
- [4] Amazon Web Services, *Amazon RDS User Guide*, AWS. - [Amazon RDS Documentation](#)
- [5] Amazon Web Services, *Amazon CloudWatch User Guide*, AWS. - [Amazon CloudWatch Documentation](#)